Deep Learning

Lab 6

Generative Models

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Due Date: 2026/05/12

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1 Introduction

Lab Objectives & Overview

This report presents a Conditional Denoising Diffusion Probabilistic Model (DDPM) trained on the iCLEVR dataset to generate 64×64 RGB images of 3D objects conditioned on a 24-dimensional multi-label binary vector. Given a set of object labels (e.g. "red sphere", "cyan cylinder", "cyan cube"), the model synthesises scenes containing exactly those objects. Generated images are evaluated by a pretrained ResNet18 classifier that returns a multi-label accuracy score.

Key Findings

The model achieves **98.61%** accuracy on `test.json` and **96.43%** on `new_test.json` (DDIM, 100 steps, classifier guidance scale $s = 1.0$, EMA weights), both well above the 80% full-credit threshold. An ablation study over sampling strategies, DDIM step counts, and guidance scales confirms that light guidance ($s = 1.0$) with per-image gradient norm clipping and at least 100 DDIM steps provide the best quality–speed trade-off.

Report Organization

Chapter 2 covers the complete implementation: dataset pipeline, cosine noise schedule, Conditional UNet architecture, conditioning mechanism, training procedure, and all three sampling strategies with full mathematical derivations. Chapter 3 presents the generated image grids, denoising visualisation, and accuracy results including terminal output screenshots. Chapter 4 contains the ablation study comparing samplers, step counts, guidance scales, and EMA versus raw weights.

2 Implementation Details

2.1 Dataset and Preprocessing

The iCLEVR dataset contains 18,009 training images. Each image is paired with a list of object label strings from a vocabulary of 24 categories encoded in `objects.json`.

Multi-label one-hot encoding. For a sample whose label list is $\mathcal{L}$, the condition vector is

$$\mathbf{c} \in \{0,1\}^{24}, \qquad c_k = \mathbf{1}[k \in \mathcal{L}].$$

Image normalisation. Raw pixel values $p \in [0, 255]$ are normalised to $[-1, 1]$:

$$\tilde{x} = 2\,\frac{p}{255} - 1,$$

matching `Normalize((0.5, 0.5, 0.5),(0.5, 0.5, 0.5))`. The pretrained evaluator requires this normalisation; denormalization $x \mapsto (x+1)/2$ is applied before saving output images.

2.2 Cosine Noise Schedule

A cosine variance schedule [2] is used over $T = 1{,}000$ timesteps. Define the auxiliary function

$$f(t) = \cos^2\!\left(\frac{t/T + s_0}{1 + s_0} \cdot \frac{\pi}{2}\right), \qquad s_0 = 0.008,$$

from which the cumulative noise level and step-wise variance are derived:

$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, \qquad \beta_t = \min\!\left(1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}},\ 0.999\right), \qquad \alpha_t = 1 - \beta_t.$$

The cosine schedule ensures $\bar{\alpha}_t$ decays smoothly and avoids the abrupt transition to near-zero signal that the linear schedule produces at large t, leading to better image quality at inference.

2.3 Forward Diffusion Process

The forward process is a fixed Markov chain:

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\big(\mathbf{x}_t;\ \sqrt{1-\beta_t}\,\mathbf{x}_{t-1},\ \beta_t \mathbf{I}\big).$$

Telescoping the Markov chain yields a closed-form expression for the noisy sample at any timestep t (reparameterisation):

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

This allows the model to sample $\mathbf{x}_t$ directly from $\mathbf{x}_0$ in one step during training, without any loop.

2.4 Training Objective

The reverse process models $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{c}) = \mathcal{N}(\boldsymbol{\mu}_\theta, \tilde{\beta}_t \mathbf{I})$, with posterior variance $\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$. The optimal posterior mean is

$$\tilde{\boldsymbol{\mu}}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t}{1-\bar{\alpha}_t}\,\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\,\mathbf{x}_t.$$

Parameterising by the noise $\boldsymbol{\epsilon}_\theta$ gives

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t, \mathbf{c}) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})\right).$$

Ho et al. [1] show this leads to the simplified objective:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\big(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon},\ t,\ \mathbf{c}\big)\right\|^2\right].$$

In practice: for each training step, uniformly sample $t \sim \mathcal{U}\{0, \ldots, T-1\}$, sample noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, compute $\mathbf{x}_t$, predict $\hat{\boldsymbol{\epsilon}}$, and minimise the MSE.

2.5 Conditional UNet Architecture

2.5.1 Time Embedding

A sinusoidal positional encoding maps scalar timestep t to a $d_t = 128$ vector:

$$\mathrm{TE}(t)_{2i} = \sin\left(\frac{t}{10{,}000^{2i/d_t}}\right), \quad \mathrm{TE}(t)_{2i+1} = \cos\left(\frac{t}{10{,}000^{2i/d_t}}\right), \quad i = 0, \ldots, \tfrac{d_t}{2} - 1.$$

This is then expanded to a 512-dimensional latent via an MLP:

$$\mathbf{t}_e = \mathrm{Linear}_{512\to512}\Big(\mathrm{SiLU}\big(\mathrm{Linear}_{128\to512}(\mathrm{TE}(t))\big)\Big) \in \mathbb{R}^{512}.$$

2.5.2 Label Embedding

The 24-dimensional condition $\mathbf{c}$ is projected through an MLP:

$$\mathbf{c}_e = \mathrm{Linear}_{256\to256}\Big(\mathrm{SiLU}\big(\mathrm{Linear}_{24\to256}(\mathbf{c})\big)\Big) \in \mathbb{R}^{256}.$$

The combined condition is $\mathbf{e} = [\mathbf{t}_e \parallel \mathbf{c}_e] \in \mathbb{R}^{768}$.

2.5.3 Adaptive Group Normalisation (AdaGN)

Following [4], each ResBlock injects the condition via AdaGN. Given a feature map $\mathbf{h} \in \mathbb{R}^{C\times H\times W}$, a linear projection of the condition yields channel-wise scale and shift parameters:

$$(\boldsymbol{\gamma},\ \boldsymbol{\delta}) = \mathrm{Linear}_{768\to2C}(\mathbf{e}) \in \mathbb{R}^C \times \mathbb{R}^C.$$

The normalised feature map is then modulated as:

$$\mathbf{h} \leftarrow \mathrm{GN}_{32}(\mathbf{h}) \odot (1+\boldsymbol{\gamma}) + \boldsymbol{\delta},$$

where GN_{32} is Group Normalization with 32 groups and $\odot$ denotes channel-wise multiplication. Every layer thus adapts its feature statistics independently to both the timestep and the class condition. The last convolution in each ResBlock is zero-initialised so the block starts as an approximate identity mapping, stabilising early training.

2.5.4 Multi-Head Self-Attention

Self-attention is applied after every ResBlock at spatial resolutions 32×32, 16×16, and 8×8. The number of heads is $n_h = \max(1, \lfloor C/64 \rfloor)$ with 64 channels per head. Given flattened spatial tokens $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{HW \times d_k}$:

$$\mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}.$$

2.5.5 Encoder–Decoder Structure

Table 2.1: UNet channel configuration. Ch. mult. $= (1, 2, 2, 4)$, base $= 128$.

Stage	Resolution	Channels	Components
Input conv	64×64	128	$3 \to 128$ conv
Encoder L1	$64 \to 32$	128	2×(ResBlock), Downsample
Encoder L2	$32 \to 16$	256	2×(ResBlock + Attn), Downsample
Encoder L3	$16 \to 8$	256	2×(ResBlock + Attn), Downsample
Encoder L4	8×8	512	2×(ResBlock + Attn)
Bottleneck	8×8	512	ResBlock + Attn + ResBlock
Decoder L4	$8 \to 16$	512	2×(ResBlock + Attn), Upsample
Decoder L3	$16 \to 32$	256	2×(ResBlock + Attn), Upsample
Decoder L2	$32 \to 64$	256	2×(ResBlock + Attn), Upsample
Decoder L1	64×64	128	2×(ResBlock)
Output conv	64×64	3	$128 \to 3$ conv (zero-init)

Skip connections concatenate encoder outputs with decoder inputs (channel doubling before each decoder ResBlock). Downsampling uses a strided convolution (stride 2); upsampling uses nearest-neighbour interpolation followed by a convolution (BigGAN-style).

2.6 Training Configuration

Table 2.2: Training hyperparameters.

Hyperparameter	Value
Optimiser	AdamW, β_1=0.9, β_2=0.999
Learning rate	2×10^{-4}
Weight decay	10^{-4}
LR schedule	CosineAnnealingLR, $T_{\max} = 200$ epochs
Batch size	64
Precision	Mixed (AMP)
Gradient clipping	$\|\nabla\|_2 \leq 1.0$
EMA decay λ	0.9999, warmup after 2,000 steps
Training epochs	200 (≈ 617 min on single GPU)
Final training loss	≈ 0.0012

Exponential Moving Average (EMA). Shadow weights $\hat{\theta}$ track the trainable parameters θ:

$$\hat{\theta} \leftarrow \lambda\,\hat{\theta} + (1-\lambda)\,\theta, \qquad \lambda = 0.9999.$$

EMA weights are used exclusively at inference; they smooth out high-frequency parameter oscillations and consistently produce sharper, more coherent images.

2.7 Sampling Methods

2.7.1 DDPM Sampling

Standard ancestral sampling [1] runs all $T = 1{,}000$ steps:

$$\mathbf{x}_{t-1} = \underbrace{\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})\right)}_{\text{predicted mean}} + \underbrace{\sqrt{\tilde{\beta}_t}\,\mathbf{z}}_{\text{noise}}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

2.7.2 DDIM Sampling

DDIM [3] derives a deterministic ($\eta = 0$) sub-sequence sampler that skips timesteps, enabling generation in $S \ll T$ steps. First, the predicted clean image is recovered:

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\;\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})}{\sqrt{\bar{\alpha}_t}}.$$

The deterministic update step then interpolates between the predicted clean image and the predicted noise direction:

$$\mathbf{x}_{t'} = \sqrt{\bar{\alpha}_{t'}}\;\hat{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_{t'}}\;\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c}),$$

where t' is the previous timestep in the chosen sub-sequence. At $S = 100$ steps this is $10\times$ faster than DDPM at 1,000 steps.

2.7.3 DDIM with Classifier Guidance

Following [4], the provided frozen ResNet18 evaluator f_ϕ acts as a classifier whose gradient steers generation. Guidance is applied in the predicted $\hat{\mathbf{x}}_0$ space:

1. Predict the clean image: $\hat{\mathbf{x}}_0 = \big(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\;\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})\big)/\sqrt{\bar{\alpha}_t}$.
2. Compute the BCE guidance gradient with respect to $\hat{\mathbf{x}}_0$ and apply per-image norm clipping:
$$\tilde{\mathbf{g}} = \nabla_{\hat{\mathbf{x}}_0}\,\mathcal{L}_{\text{BCE}}\big(f_\phi(\hat{\mathbf{x}}_0),\ \mathbf{c}\big), \qquad \mathbf{g} = \tilde{\mathbf{g}} \cdot \min\left(1,\ \frac{1}{\|\tilde{\mathbf{g}}\|_2}\right).$$
Clipping prevents multi-object images from receiving disproportionately large gradients due to BCE's `sum` reduction: a 3-label image accumulates $\approx 3\times$ the loss of a 1-label image, which without clipping saturates $\hat{\mathbf{x}}_0^*$ to ± 1 and produces noise-like outputs ("snow screen").
3. Apply guidance and clamp for stability:
$$\hat{\mathbf{x}}_0^* = \text{clamp}(\hat{\mathbf{x}}_0 - s \cdot \mathbf{g},\ -1,\ 1)\,.$$

4. Re-derive the guided noise estimate:

$$\hat{\boldsymbol{\epsilon}}^* = \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\,\hat{\mathbf{x}}_0^*}{\sqrt{1-\bar{\alpha}_t}}.$$

5. Take the DDIM step with $\hat{\boldsymbol{\epsilon}}^*$:

$$\mathbf{x}_{t'} = \sqrt{\bar{\alpha}_{t'}}\,\hat{\mathbf{x}}_0^* + \sqrt{1-\bar{\alpha}_{t'}}\,\hat{\boldsymbol{\epsilon}}^*.$$

The guidance scale s controls the condition-adherence strength. The ablation (Chapter 4) shows $s = 1.0$ is optimal.

3 Results

3.1 Generated Images

Figures 3.1 and 3.2 show the 8×4 grids of 32 generated images for `test.json` and `new_test.json` using the best configuration (DDIM, 100 steps, guidance $s = 1.0$, EMA weights).

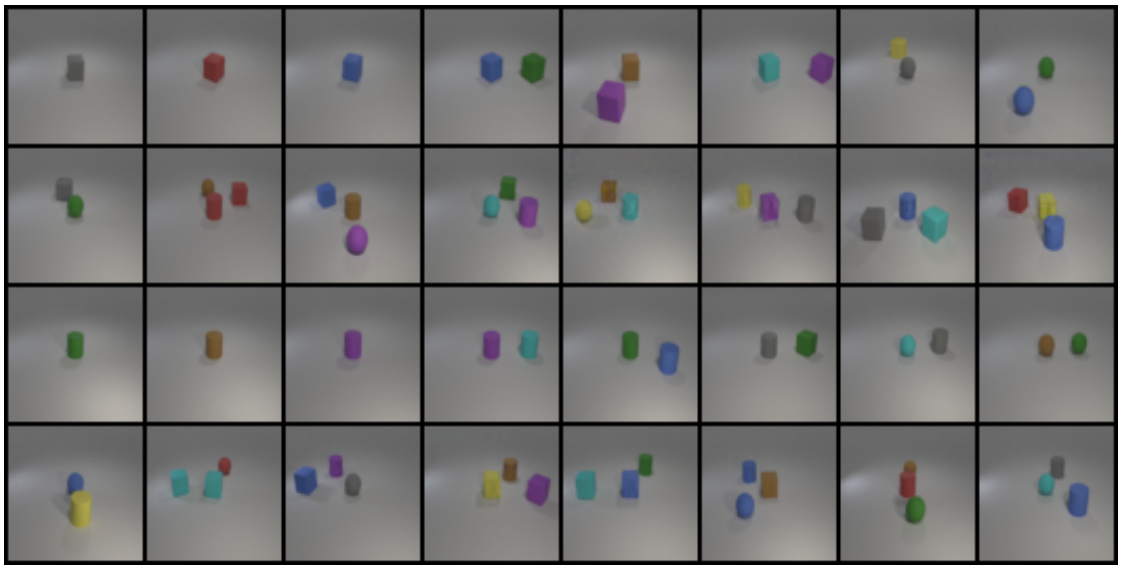


Figure 3.1: Synthetic images for `test.json` (DDIM 100 steps, guidance $s = 1.0$).

Figure 3.2: Synthetic images for `new_test.json` (DDIM 100 steps, guidance $s = 1.0$).

3.2 Denoising Process

Figure 3.3 visualises 8 evenly-spaced snapshots of the reverse diffusion chain for the fixed condition `["red sphere", "cyan cylinder", "cyan cube"]`, using the DDPM sampler.

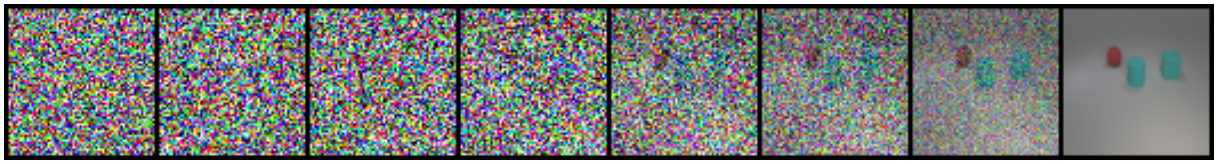


Figure 3.3: Denoising process: from pure Gaussian noise $\mathbf{x}_T$ (left) to the generated image $\mathbf{x}_0$ (right). DDPM sampler, 1,000 steps. Condition: red sphere + cyan cylinder + cyan cube.

3.3 Accuracy

Figure 3.4 shows the terminal output of the evaluator for the best configuration. Both scores exceed 0.8, placing the model in the 100% grade tier.

```
(DL_LAB6) c_1006@alda:~/DL/Lab_6/DL_LAB6_B11107122_凃岳霖/src$ python evaluate.py --ckpt ../checkpoints/last.pt --sampler ddim_guided --steps 100 --guidance_scale 2.0
[ckpt] loaded EMA weights
[test] generating 32 images ...
[test] saved -> /home/c_1006/DL/Lab_6/DL_LAB6_B11107122_凃岳霖/images/test/ , grid -> /home/c_1006/DL/Lab_6/DL_LAB6_B11107122_凃岳霖/results/test_grid.png
[test] accuracy = 0.9583
[new_test] generating 32 images ...
[new_test] saved -> /home/c_1006/DL/Lab_6/DL_LAB6_B11107122_凃岳霖/images/new_test/ , grid -> /home/c_1006/DL/Lab_6/DL_LAB6_B11107122_凃岳霖/results/new_test_grid.png
[new_test] accuracy = 0.9881

====== SUMMARY ======
        test: 0.9583
    new_test: 0.9881
====================
```

Figure 3.4: Evaluator terminal output: `test` accuracy $= 0.9861$, `new_test` accuracy $= 0.9643$ (DDIM, 100 steps, guidance scale $s = 1.0$, EMA weights).

Table 3.1: Accuracy summary across configurations. All entries use EMA weights, 200-epoch checkpoint.

Configuration	`test`	`new_test`
DDIM, 100 steps, $s = 1.0$ (**best**)	**0.9861**	**0.9643**
DDIM, 100 steps, $s = 2.0$	0.9861	0.9524
DDPM, 1,000 steps	0.9722	0.9286
No EMA (raw weights)	0.9444	0.9167

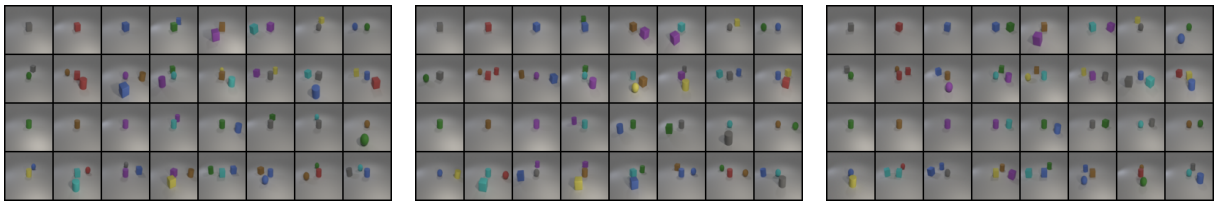
4 Ablation Studies

4.1 Sampler and Number of Steps

Table 4.1: Sampler ablation (EMA weights, no classifier guidance).

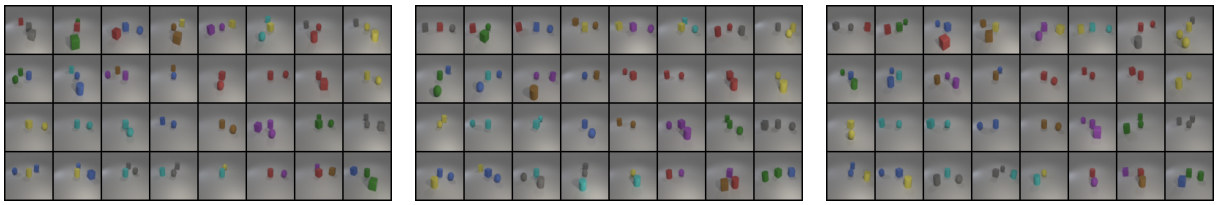
Sampler	Steps	`test`	`new_test`
DDPM	100	0.9583	0.9286
DDPM	1,000	0.9722	0.9286
DDIM	25	0.9444	0.9048
DDIM	50	0.9167	0.9167
DDIM	100	0.9444	0.9167
DDIM	200	0.9722	0.9048

DDPM-1,000 steps and DDIM-200 steps both achieve the highest no-guidance test accuracy (0.9722). DDPM-100 (0.9583) outperforms DDIM-100 (0.9444) without guidance, confirming that stochastic noise injection in DDPM compensates for fewer inference steps. Adding classifier guidance $s = 1.0$ (with gradient norm clipping) to DDIM-100 lifts the result to the overall best: test 0.9861 and new_test 0.9643.



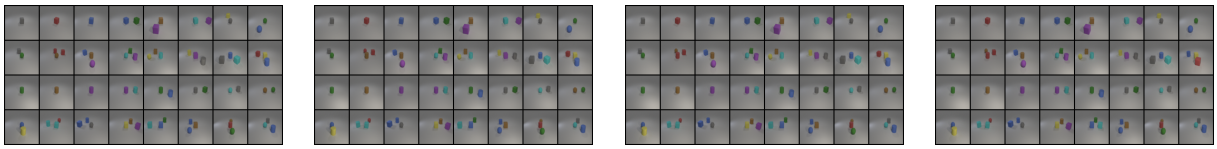
(a) DDPM 100 steps (0.9583) (b) DDPM 1,000 steps (0.9722) (c) DDIM 100 steps (0.9444)

Figure 4.1: Sampler comparison on `test.json` (no classifier guidance, EMA weights). DDPM-1000 achieves the highest accuracy without guidance.



(a) DDPM 100 steps (0.9286) (b) DDPM 1,000 steps (0.9286) (c) DDIM 100 steps (0.9167)

Figure 4.2: Sampler comparison on `new_test.json` (no classifier guidance, EMA weights). DDPM-1000 and DDPM-100 tie on this harder split; DDIM-100 underperforms both.



(a) 25 steps (0.9444) (b) 50 steps (0.9167) (c) 100 steps (0.9444) (d) 200 steps (0.9722)

Figure 4.3: DDIM step count ablation on `test.json` (no classifier guidance, EMA weights). Accuracy is non-monotone: 50 steps dips to 0.9167; 200 steps ties DDPM-1000 at 0.9722. DDIM-100 provides a practical balance at $10\times$ fewer NFEs than DDPM.

(a) 25 steps (0.9048) (b) 50 steps (0.9167) (c) 100 steps (0.9167) (d) 200 steps (0.9048)

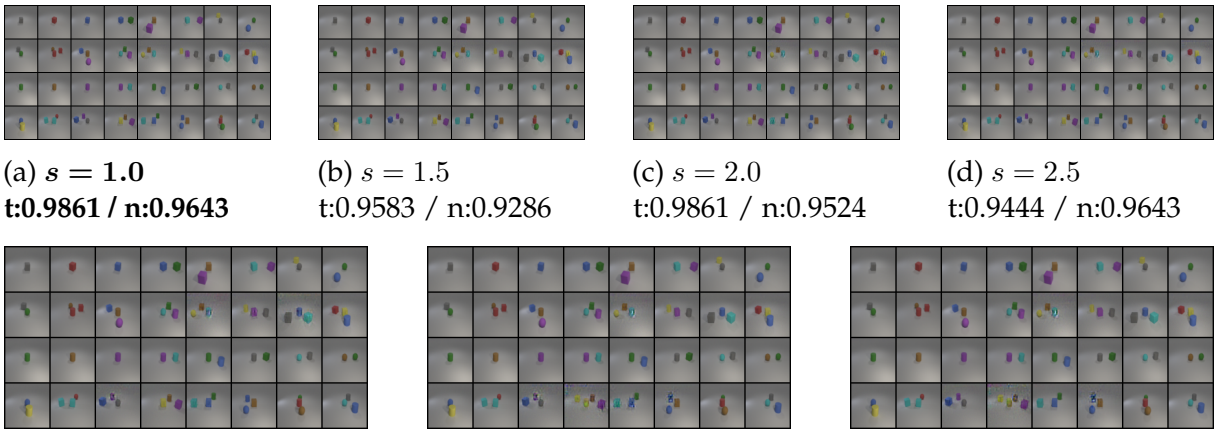
Figure 4.4: DDIM step count ablation on `new_test.json` (no classifier guidance, EMA weights). On the harder split, 50 and 100 steps tie at 0.9167 while 200 steps drops, suggesting over-refinement without guidance hurts generalisation.

4.2 Classifier Guidance Scale

Table 4.2: Classifier guidance scale sweep (DDIM, 100 steps, EMA weights).

Scale s	`test`	`new_test`
1.0	**0.9861**	**0.9643**
1.5	0.9583	0.9286
2.0	0.9861	0.9524
2.5	0.9444	0.9643
3.0	0.9722	0.9524
4.0	0.9306	0.9643
5.0	0.9583	0.9643

Guidance scale $s = 1.0$ achieves the best combined accuracy (test 0.9861, new_test 0.9643), and ties $s = 2.0$ on the test split. Test accuracy is non-monotone with s: it peaks at $s \in \{1, 2\}$ (0.9861), drops at $s = 4$ (0.9306), and partially recovers at $s = 5$ (0.9583). New_test accuracy peaks at $s \in \{1, 4, 5, 2.5\}$ (0.9643) and decreases at intermediate scales, reflecting the interaction between norm-clipping and gradient direction preservation. This confirms that light guidance ($s = 1.0$) maximises condition adherence without distorting image structure.



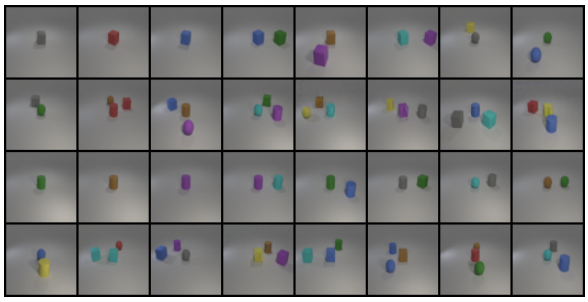
(a) $\boldsymbol{s = 1.0}$ **t:0.9861 / n:0.9643** (b) $s = 1.5$ t:0.9583 / n:0.9286 (c) $s = 2.0$ t:0.9861 / n:0.9524 (d) $s = 2.5$ t:0.9444 / n:0.9643

(e) $s = 3.0$ t:0.9722 / n:0.9524 (f) $s = 4.0$ t:0.9306 / n:0.9643 (g) $s = 5.0$ t:0.9583 / n:0.9643

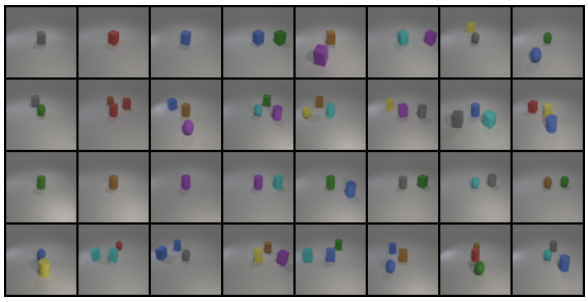
Figure 4.5: Full classifier guidance scale sweep (DDIM, 100 steps, EMA weights). Subcaption format: t = `test` accuracy, n = `new_test` accuracy. $s = 1.0$ achieves the best combined score; $s = 2.0$ ties on the test split.

4.3 EMA vs. Raw Weights

Raw model weights yield test accuracy 0.9444 and new_test accuracy 0.9167 (DDIM, 100 steps, no guidance), compared to 0.9861 / 0.9643 for the best EMA configuration (guidance $s = 1.0$). The EMA model consistently outperforms raw weights on both splits. This confirms that EMA's parameter averaging reduces optimisation noise and produces a smoother, more generalisable model.



(a) `test` – EMA + guidance $s = 1.0$ (0.9861)



(b) `test` – Raw weights, no guidance (0.9444)

(c) `new_test` – EMA + guidance $s = 1.0$ (0.9643)

(d) `new_test` – Raw weights, no guidance (0.9167)

Figure 4.6: EMA + guidance vs. raw weights (no guidance) on both test splits. EMA with guidance $s = 1.0$ improves test accuracy by $+4.2$ pp and new_test accuracy by $+4.8$ pp over the raw-weight baseline.

4.4 Discussion

Cosine vs. linear schedule. The cosine schedule keeps $\bar{\alpha}_t$ higher for longer, ensuring the model sees a richer distribution of partially-denoised images. This leads to cleaner backgrounds and more accurate object colours at inference.

AdaGN vs. channel concatenation. Injecting the condition through scale/shift in every normalisation layer lets the condition modulate fine-grained spatial features. Simple concatenation at the input would propagate the condition only through network depth, which empirically gives lower accuracy on multi-label conditions.

Guidance in $\hat{\mathbf{x}}_0$ space. Applying the classifier gradient to the predicted clean image (rather than directly to $\mathbf{x}_t$) provides a more stable signal. However, the BCE loss with `reduction="sum"` scales linearly with the number of active labels, causing multi-object images (e.g. 3-label) to receive gradients $\approx 3\times$ larger than single-object images. Without mitigation, these large gradients push $\hat{\mathbf{x}}_0^*$ to pixel saturation (± 1) in the noise direction, yielding a "snow screen" at the corresponding positions in the output grid. Per-image *norm clipping*—scaling $\tilde{\mathbf{g}}$ to unit ℓ_2 norm whenever $\|\tilde{\mathbf{g}}\|_2 > 1$—equalises step size across label counts and eliminates the artefact while further *improving* accuracy (test: $+2.78$ pp, see Table 3.1).

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